

Session:-3

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

ans:-

- HTTP (Hypertext Transfer Protocol): Used to fetch plain HTML pages and transmit web resources between a browser and a web server.
- HTTPS (Hypertext Transfer Protocol Secure): An encrypted version of HTTP used to securely transmit sensitive data over the internet.
- FTP (File Transfer Protocol): Used for transferring files directly between a client and a remote server over a network.
- SMTP (Simple Mail Transfer Protocol): Used to send and route outbound email messages across networks.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

ans:-

- IPv4 uses a 32-bit address format providing about 4.3 billion unique IP addresses.
- IPv6 uses a 128-bit hexadecimal format to provide a virtually limitless pool of unique IP addresses.

3. Run nslookup google.com on your system and write down the IP address you get.

ans:-

- nslookup google.com Output
- Command: nslookup google.com
- Sample IP Address Output: 142.250.190.46

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

ans:-

- 200 (OK): The request succeeded, and the server returned the requested data.
- 301 (Moved Permanently): The requested page/resource has been permanently moved to a new URL.
- 404 (Not Found): The server cannot find the requested URL or resource.
- 500 (Internal Server Error): The server encountered an unexpected error and could not fulfill the request.
- 403 (Forbidden): The server understands the request but refuses to authorize access.

5.Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

ans:-

- DNS Hierarchy Example (www.google.com)
- Root (.): The invisible dot at the end (www.google.com.) managed by root servers, directing queries to the top-level directory.
- TLD (Top-Level Domain - .com): The main extension category managed by generic TLD registries.
- Domain (google): The primary registered name/brand identity under the TLD.
- Subdomain (www): A specific sub-section or service defined under the main domain.